

Solution Requirements

Date	14 july 2026
Team ID	xxxxxxx
Project Name	opticrop
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system shall provide secure user authentication using username and password. Passwords shall be securely hashed before storage, and only authenticated users can access the application.</i>	High
2	Authorization levels	<i>The system shall support two roles: Admin and Farmer/User. Admins can manage users, datasets, crop models, and prediction records. Farmers can register, log in, request crop</i>	High

		<i>recommendations, and view their prediction history.</i>	
3	External interfaces	<i>The system shall communicate with the trained Machine Learning model, SQLite database, and Flask REST APIs. Future versions may integrate weather APIs to obtain real-time environmental data.</i>	Medium
4	Transactions processing	<i>The system shall generate crop prediction reports, user activity reports, recommendation history, and administrative dashboards showing prediction statistics and system usage</i>	High
5	Reporting	<i>Crop recommendations shall be generated only when all required parameters (Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall) are provided. Invalid or incomplete inputs shall be rejected with appropriate validation messages.</i>	Medium
6	Business rules, etc.	<i>The system shall protect user information by following standard data privacy and cybersecurity practices. Personal information shall not be disclosed without authorization.</i>	High

7	Compliance to laws or regulations	<i>The system shall protect user data in accordance with applicable data privacy regulations and ensure secure storage of user credentials.</i>	Medium
8	Other	<i>The system shall provide confidence scores for crop predictions, maintain prediction logs, and support future model updates without affecting the user interface. High.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	<i>The system shall provide crop recommendations quickly with minimal delay.</i>	<i>Prediction response time < 2 seconds under normal load.</i>
2	Scalability	<i>The system shall support increasing numbers of users and larger agricultural datasets without significant performance degradation.</i>	<i>Supports 500+ concurrent users and scalable database growth.</i>
3	Security & Data Privacy	<i>User credentials shall be encrypted, API endpoints protected, and input validation performed to prevent attacks such as SQL injection and XSS.</i>	<i>Password hashing, HTTPS support, secure authentication, and protected APIs.</i>

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

4	Reliability & Availability	<i>The application shall remain available with minimal downtime and provide continuous crop recommendation services.</i>	<i>99.9% monthly uptime with automatic error logging and recovery.</i>
5	Usability & Accessibility	<i>The user interface shall be simple, responsive, and easy to use for both farmers and administrators across desktop and mobile devices.</i>	<i>Responsive UI with intuitive navigation and cross-browser compatibility.</i>
6	Other	<i>The system shall use a modular architecture to simplify maintenance, testing, and deployment across different platforms. Docker support shall enable easy deployment.</i>	Modular codebase, Docker deployment, and complete technical documentation.